

Towards an Active Inference Personality Framework: The Deep-Self Predictive Cascade Model

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Inter-individual differences in personality currently suffer from a weak generative and mechanistic conceptualisation. By bridging active inference frameworks with established personality accounts, we introduce a principled hierarchical predictive architecture grounded in the Free Energy Principle. In contrast to descriptive psychometric frameworks that map behavioural phenotypes, the Deep-Self Predictive Cascade Model proposes that observable personality traits are the downstream expressions of deep prior expectations and precision biases that shape how an agent interprets uncertainty, regulates action, and maintains a structural sense of self. Within this architecture, traits are conceptualised not as static endowments, but as stable attractor basins within a Free Energy landscape. The hierarchy cascades from foundational ultra-priors governing epistemic, allostatic, and teleological expectations, through physiological affective states, and into the precision-weighting of the minimal, agentic, and narrative self-models. By parameterising these computational layers, the Deep-Self Model naturally generates the phenotypic profiles described by the Big Five, Cybernetic Big Five Theory, BIS/BAS, and the HiTOP dimensional taxonomy of psychopathology. This framework offers a unified computational ontology for personality, yielding testable predictions for cognitive neuroscience, psychophysiology, and computational psychiatry.

Impact Statement

The Deep-Self Predictive Cascade Model provides a mechanistic, mathematically grounded account of how hierarchical expectations and precision biases generate the full range of observable personality phenotypes, offering a principled bridge between the Free Energy Principle and the scientific study of individual differences.

Keywords: active inference, free energy principle, personality, individual differences, predictive processing, cybernetics, computational psychiatry

Introduction

Personality psychology currently faces a persistent explanatory gap. Descriptive frameworks, most notably the Five-Factor Model (Goldberg, 1990; McCrae & John, 1992), have proven remark-

ably successful at organising individual differences in behaviour, affect, and cognition. However, their structural parsimony comes at a theoretical cost. These models describe phenotypic variance across populations but offer limited mechanistic accounts

of the generative processes that produce and maintain such variation within the individual.

Several frameworks have attempted to bridge this descriptive-process divide. The Cognitive-Affective Personality System (CAPS, [Mischel & Shoda, 1995](#)) conceptualises personality as a network of interacting cognitive and affective units, while Cybernetic Big Five Theory (CB5T, [DeYoung, 2015](#)) offers a functionalist account of traits as the typical parameters of a cybernetic system optimising for distinct goals. Yet, while these models successfully introduce process dynamics, they lack a unified mathematical ontology to explain *why* certain cognitive-affective parameters reliably coalesce into stable phenotypes.

The Free Energy Principle (FEP, [Friston, 2010](#)) and its corollary process theory, active inference, offer a principled candidate for such an ontology. Under the FEP, self-organising biological systems maintain their structural integrity by minimising the upper bound on their own surprise—formally defined as variational free energy. This is achieved by continuously updating probabilistic generative models of their environment via perceptual inference (revising internal models) and active inference (acting upon the world).

Applied to individual differences, this implies that personality characteristics are not fixed endowments. Rather, they are the downstream behavioural and affective consequences of deeply entrenched probabilistic parameters. To formalise this, we introduce the **Deep-Self Predictive Cas-**

cade Model. This model posits that personality is fundamentally a *Free Energy Landscape*—a topographic surface mapping achievable parameter configurations against their associated expected free energy.

The binary distinction between expectations and biases clarifies this topology: **Expectations (location parameters, μ)** determine *where* the attractor basins are carved into the landscape, establishing the underlying assumptions the agent holds about reality. **Biases (precision weightings, π)** determine *how deep and steep* those basins are, defining how easily the system can be dislodged from its habitual states by sensory noise. In this view, traits are not things we have; they are the attractors we reliably fall into.

The Deep-Self Predictive Cascade Model

The Deep-Self Model is structured as a hierarchical cascade comprising four distinct computational levels. Information propagates downstream, wherein the precision of higher-level prior expectations acts as a multiplier on their downward influence, structurally dictating the parameters of the lower levels.

Level 1: Ultra-Priors

At the apex of the hierarchy are “Ultra-Priors”—the system’s unconditioned, foundational expectations regarding the structure of reality and the agent’s capacity to navigate it. These occupy the absolute apex of the cascade, acting as hyperparameters unconstrained by higher psychological structures.

The Epistemic Imperative

This imperative drives the resolution and management of uncertainty, dictating how the generative model handles world-opacity. * **Volatility:** The baseline expectation regarding the rate of environmental change (the ω parameter in Hierarchical Gaussian Filtering; Mathys (2011)). High expected volatility heavily discounts past learning and braces

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the system for persistent prediction errors, serving as the primary upstream driver of epistemic anxiety.

* **Noise:** The expectation of irreducible, aleatoric uncertainty. This corresponds to the ambient statistical “floor” of noise the agent expects to tolerate. High expected noise represents an epistemic humility that prevents premature cognitive closure, whereas a low noise expectation triggers drastic, often maladaptive model updates when residual variance cannot be eliminated.

The Allostatic Imperative

This constitutes the pragmatic, metabolic drive to maintain systemic integrity. * **Energy:** The expectation of allostatic reserve. Grounded in predictive processing accounts of the “body-budget” (Barrett, 2017), this parameter dictates whether the agent expects to operate at a metabolic surplus or a chronic deficit. * **Tractability:** The expectation that active inference can reliably reduce prediction error. Formally related to the expected precision of proprioceptive and control signals, tractability determines whether the agent perceives the environment as controllable. A low tractability expectation shares close computational parallels with learned helplessness (Seligman, 1972), driving epistemic shielding and behavioural inhibition as energetically efficient responses to expected futility.

The Teleological Imperative

This imperative governs goal-directed purposiveness and orientation toward finitude. * **Reward:** The expectation concerning the value of available outcomes. Formally corresponding to the log prior probability over preferred outcomes (the **C**-vector in active inference), it anchors approach motivation by weighting expected free energy toward intrinsically valuable states. * **Horizon (The Terminal Expectation):** The expected temporal depth of the agent’s policy space (parameter T). Within the FEP, mortality salience functions as a precision-weighting problem: confrontation with finitude forces the agent to resolve the prediction error of its anticipated systemic cessation (Greenberg et

al., 1986). This defines whether the agent aims for *Transcendence* (extending the Markov blanket across time through cultural or biological legacy), *Integration*, or *Disengagement* (collapsing temporal depth to avoid the massive free energy of an uncontrollable future, which, in its extreme, functions as the active inference analogue of Thanatos).

Level 2: Affective States

The second level represents the physiological and affective readouts of Level 1 priors. Deviating from classical discrete emotion theories, the Deep-Self Model aligns with the theory of constructed emotion (Barrett, 2017), treating affect as the phenomenological readout of predictive success or failure.

- **Epistemic Arousal:** The felt interoceptive consequence of unresolved, high-precision prediction errors. Driven by high Volatility, chronic epistemic arousal forces the system to retreat to its most reliable data stream—the physical body—resulting in hyper-vigilance.
- **Vitality:** The somatic readout of the body-budget. A high Vitality state signals surplus allostatic resources, facilitating structurally demanding behaviours like social coupling and exploration. Conversely, low Vitality reflects allostatic load.
- **Valence:** The affective readout tracking the system’s own Free Energy derivative. Modulated directly by Reward Expectation and constrained by Vitality, it represents anticipatory positive affect and the baseline Expected Free Energy tone of the system.

Level 3: Precision Biases

Affective states cascade downward to tune Level 3 Precision Biases. This level forms the structural architecture of the self, mapping cleanly onto phenomenological taxonomies of selfhood.

The Minimal Self

Anchoring the physiological boundary of the Markov blanket, governed heavily by interoceptive circuitry (Seth, 2013). * **Interoception-Exteroception Bias:** Dictates reliance on internal somatic signals versus external structure. * **Differentiation-Dissolution Bias:** Governs the permeability of the Markov blanket. *Differentiation* favours autonomy and rigid boundaries, motivating dominance-seeking to subjugate the environment rather than revise the internal model. *Dissolution* favours structural synchrony, driving affiliation and the distribution of cognitive load across the social group.

The Agentic Self

Representing the cybernetic strategy for resolving expected free energy. * **Action-Perception Bias:** The relative precision placed on active inference (modifying the world) versus perceptual inference (modifying internal beliefs). * **Exploration-Exploitation Bias:** The precision assigned to epistemic foraging. Crucially, this drive is split into two distinct operational domains: **Somatic Foraging** (testing the physical Markov blanket through biological edge-seeking and physical risk) and **Semantic Foraging** (testing the conceptual boundaries of the generative model by probing worldview coherence through transgressive art or profound altered states of consciousness).

The Narrative Self

Governing diachronic (temporal) continuity. * **Rigidity-Fluidity Bias:** The precision assigned to autobiographical stability. High rigidity conserves identity defensively across contexts; fluidity allows for rapid self-updating but introduces the risk of narrative fragmentation (identity diffusion). * **Future-Present Bias:** A temporal discounting bias determining how quickly the precision of predicted outcomes degrades over time.

Level 4: Phenotypic Attractors (Observables)

The confluence of Level 3 precision biases creates the stable attractor basins that traditional psy-

chometrics classify as traits. By parameterising these deep computational layers, the Cascade Model natively generates the phenotypic profiles spanning several major psychological frameworks.

Within the **Five-Factor space**, *Neuroticism* emerges not as a fundamental trait, but as an attractor generated by chronically elevated epistemic arousal feeding a heavy Interoception bias. *Extraversion* manifests through converging precision on the Action pole and the Dissolution pole. *Openness* requires a highly fluid Narrative Self paired with an Exploration bias.

Similarly, the dimensions of **Attachment Theory** reflect the Differentiation-Dissolution bias constrained by Level 1 priors. *Anxious attachment* emerges from high Epistemic Arousal driving hyper-coupling (Dissolution), where the agent urgently uses another's Markov blanket for co-regulation but lacks the tractability to trust the relationship. *Avoidant attachment* is driven by low Tractability or high metabolic cost, prompting defensive Differentiation and withdrawal.

Relationship to Deep Structures and Interventions

Jungian Archetypes & The REBUS Model

The topographic nature of the cascade accommodates deep phylogenetic structures. Jungian archetypes can be formalised as deeply entrenched, phylogenetically conserved structural expectations acting as “strange attractors” within the free energy landscape (Becker & Neuberg, 2019). Furthermore, the model aligns smoothly with the REBUS (Relaxed Beliefs Under Psychedelics) framework (Carhart-Harris & Friston, 2019). The therapeutic benefits of extreme semantic foraging via psychedelics can be understood as the functional consequence of transient attractor-wall shallowing—liberating the agent from pathologically stable configurations and allowing the system to settle into adjacent basins with lower expected free energy.

Limitations and Open Questions

The most pressing empirical challenge for the Deep-Self Model is developing valid measurement instruments capable of assessing the proposed expectation and bias parameters. Traditional subjective questionnaires predominantly capture the downstream “observables” of personality; they do not directly access the unconscious precision weightings that form the actual predictive scaffolding of experience.

Because these deeply entrenched generative parameters operate outside declarative awareness, implicit behavioural and psychophysical tasks offer a much more promising path forward. Paradigms investigating illusion sensitivity and phenomenological control are particularly well-suited to probe the underlying predictive architecture directly. Future research should leverage machine psychophysics and computational modelling to reliably back-estimate these deep, sub-personal parameters from task performance, providing rigorous empirical validation for the cascade hierarchy.

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